

The 10th International Conference on Computer Science and Computational Intelligence 2025

Comparative analysis of two-class and multi-class toxicity detection using multi-source gaming chat data

Kasandika Andariefli^{a,*}, Jervino Leonard^a, Vincentius Dewanto^a, Andien Dwi Novika^a

^a Computer Science Department, School of Computer Science, Bina Nusantara University, Jakarta, Indonesia 11480

Abstract

Toxic chat remains a significant problem in competitive online games like Dota 2, where abusive language can degrade player experience and disrupt community health. This study focuses on detecting such messages by training deep learning models on a merged dataset from Kaggle and Hugging Face, combining chat logs labelled for toxicity. We evaluated two architectures: a Long Short-Term Memory (LSTM) model and a Convolutional Neural Network (CNN) with a 1D convolutional layer and GlobalMaxPooling. Results showed that the LSTM model struggled, predicting only the majority non-toxic class, which led to a misleading overall test accuracy of 53.4% and zero scores for the minority classes. In contrast, the CNN achieved an overall accuracy of 79.9% and demonstrated balanced performance, with F1-scores of 0.64 and 0.66 for toxic and severe-toxic classes, respectively. These results highlight CNN's ability to capture local linguistic patterns in short chat messages, despite the dataset's significant class imbalance. Our findings emphasize the importance of addressing data imbalance in toxicity detection and selecting model architectures suited for short, dynamic gaming chat environments. These insights can inform the development of robust moderation tools in online gaming communities, promoting healthier and more inclusive interactions. Future research will explore integrating transformer-based models and data balancing techniques to further improve detection accuracy.

© 2025 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the 10th International Conference on Computer Science and Computational Intelligence (ICCSCI) 2025

Keywords: Toxic Chat Detection; Deep Learning; Neural Network

* Corresponding author. Tel.: +62-812-988-8000.

Email Address: kasandika.andariefli@binus.ac.id

1. Introduction

Toxicity in online messages refers to harmful, offensive, or abusive content that can negatively impact communication and community health. It encompasses a range of behaviours, from explicit insults to more subtle forms of harassment [1]. In gaming environments, research highlights the varied manifestations and impacts of toxicity across different game genres. For example, in team-based competitive games such as League of Legends, toxic behaviours include intentional feeding, verbal abuse, and grieving [2]. In Valorant, female players experience gender-based harassment and verbal abuse, leading to reluctance in using voice communication [3]. The StarCraft II community exhibits ten categories of toxic behaviour, ranging from mildly annoying to severely traumatizing [4]. These toxic behaviours negatively affect players well-being, with studies showing associations between perceived toxicity and lower in game social capital, reduced satisfaction of relatedness needs, and increased loneliness [5]. Toxic victimhood is encouraged by low self-efficacy and high online disinhibition, whereas normalization of toxicity as a part of competitive gaming, furthered by conduct reconstruction and dehumanization along with cultural context, exacerbates the situation due to players being unwilling to report it [6], [7], [8].

Toxicity detection is essential in maintaining healthy interaction, especially on open-source software communities, with prior researchers finding that both an individual user propensity and the discussion context are effective predictors of toxic language [9], [10]. Traditional computer-based solutions to combat toxicity in gaming uses simple computer-based features such as in-game audio features and metadata to predict toxic behaviour with high accuracy [11]. As text-based communication became more prominent, researchers began employing text mining techniques to analyse chat logs for sexist or abusive language, with some using Naïve Bayes classifiers to identify patterns of toxicity [12]. In more recent research, neural networks have been applied to text mining tasks, including classification, categorization, and summarization. These efforts have laid the groundwork for the adoption of more advanced language-aware models.

Architectures such as LSTM and Bi-LSTM have demonstrated high accuracy in detecting various forms of toxicity, including threats, insults, and hate speech, with Bi-LSTM consistently outperforming standard LSTM models and achieving accuracies ranging from 94.94% to 98.07% [13], [14], [15]. Convolutional Neural Networks have also been proven effective such as an ensemble of CNN and NLP, AlexNet-Inspired 3-Tier CNN or even multichannel CNN, these models have demonstrated high performance, with reported accuracies ranging from 98.05% to 98.505% and ROC-AUC scores of 0.9846 to 0.9854 [16], [17], [18]. Transformers have emerged as powerful tools in text mining, demonstrating effectiveness across various tasks. In supervised settings, pre-trained Transformer-based word embeddings significantly improve performance in text classification, named entity recognition, and question answering. Ait-Saada et al. reported that BERT-base and RoBERTa-base models achieved strong performance in estimating the number of clusters across multiple datasets, with normalized mutual information (NMI) scores of 0.98 for classic3 and 0.74 for classic4, as well as an NMI of 0.75 for the BBC dataset. RoBERTa-large also demonstrated high NMI performance, reaching 0.75 for the BBC dataset, alongside 0.98 for classic3 and 0.74 for classic4. These results underscore the superior ability of transformer-based architectures to capture semantic nuances and structural patterns in text, reinforcing their suitability for complex natural language processing tasks [19]. Among these, BERT has emerged as a particularly influential model in NLP research and applications.

Originally developed by Google, BERT (Bidirectional Encoder Representations from Transformers) leverages transformer encoders and masked language modelling to pretrain on massive unlabelled corpora. Its deeply bidirectional architecture and transfer learning capabilities have revolutionized many NLP tasks. Pretrained on BookCorpus and English Wikipedia, BERT has been fine-tuned for eleven GLUE benchmark tasks, achieving state-of-the-art results. For gaming's toxicity detection, the contextual understanding of BERT and its robustness for data imbalance strengthens its potential as a candidate for future enhancements.. Prior studies have shown that fine-tuning BERT on small datasets (100 to 2,500 texts) can outperform classical models by 10.7 to 18.3 percentage points across various tasks. It has also achieved performance comparable to models trained on 5,000 texts using only 500 and has proven effective on imbalanced data [20], [21]. Additionally, BERT-based models have been successfully applied to spam detection and sentiment analysis, achieving F1-scores of up to 91% and 88.94% respectively [22], [23]. Transfer learning with BERT also offers efficiency benefits, reducing training time by 30 to 50 percent while maintaining high accuracy across dataset sizes [24].

While CNNs, LSTMs, and BERT-based models have all shown strong performance in toxicity detection, few studies have investigated how their performance varies under different classification schemes, particularly in gaming-specific contexts. The choice between binary and multi-class classification presents a critical trade-off, that being binary setups offer simplicity and often yield higher accuracy, while multi-class models provide finer-grained insights into the type of severity in toxicity. Prior research in sentiment analysis and multi-choice QA has shown that binary models can outperform multi-class ones by 0.85% to 5%, with binary classification reaching up to 80.0% accuracy compared to 74.7% for multi-class models [25], [26], [27], [28]. Despite these findings, there is limited exploration of this trade-off in the domain of short, dynamic in-game chats. Our study fills this gap by comparing CNN, LSTM, and BERT architectures under both schemes, offering new insights into how classification granularity interacts with model complexity in the context of real-world gaming toxicity [29], [30].

2. Methodology

2.1 Dataset Composition

The study utilizes a combined dataset derived from two publicly available sources:

1. GOSU AI English Dota 2 Game Chats:
 - a. Source: A kaggle dataset curated by Daniel Fesalbon [31].
 - b. Content: An extensive collection of English-language in-game chat messages from Dota 2 matches.
 - c. Structure: Each entry comprises a text field containing the chat message and a label indicating its toxicity level.
2. Dota 2 Toxic Chat Data:
 - a. Source: Hugging Face dataset provided by dffesalbon [32].
 - b. Content: A curated set of Dota 2 in-game chat messages labelled for toxicity.
 - c. Structure: Each entry includes a text field and a corresponding label denoting the toxicity classification.

Table 1. Sample dataset after the two datasets are merged

Message	Target
why u guys so smell and poor country	2
i thought we were friends	0
he is best shittalker au	1

The two datasets were merged to form a comprehensive corpus for training and evaluation. From the Kaggle dataset, only the text and label columns were extracted to maintain consistency with the Hugging Face dataset, which inherently contains only these two fields. The combined dataset ensures a diverse representation of in-game communications, capturing various expressions of player interactions.

Table 2. Distribution of Toxicity Classes in the Combined Dataset

Class	Description	Number of Instances
0	Non-Toxic	2,987
1	Toxic	926
2	Severe-Toxic	1,556

To better illustrate the data imbalance, Table 2 presents the distribution of toxicity classes in the combined dataset. It shows a significant skew towards the non-toxic class and highlights the challenges faced by models when

learning to detect minority toxic classes.

2.2 Text Preprocessing and Word Embedding

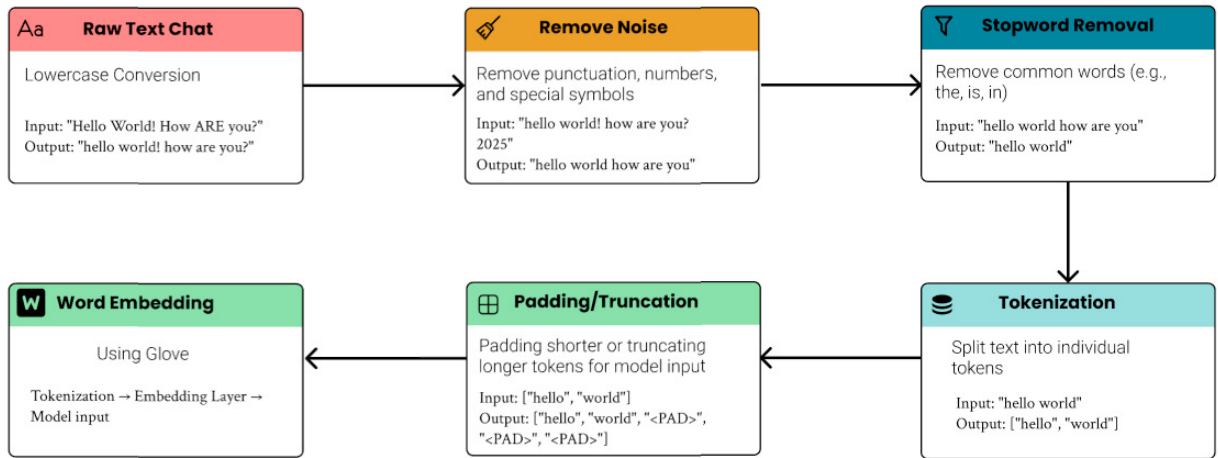


Fig. 1. Workflow of Data Preprocessing

To standardize the text data for the models to effectively learn patterns:

1. Lowercasing: All texts are converted to lowercase.
2. Noise Removal: removal of punctuations marks, special symbols or characters, and numbers, all to reduce necessary data.
3. Stopword Elimination: Using the common English stop words NLTK library
4. Tokenization: Sentences were tokenized into individual words using Keras' Tokenizer.
5. Padding: Token sequences were padded or truncated to a fixed length of 150 tokens to maintain consistent input dimensions for the model.

The toxicity labels from both datasets were standardized into a unified numerical encoding scheme to enable consistent model training. The following mapping was applied:

- 0 – non-toxic
- 1 – toxic
- 2 – severe-toxic

After text preprocessing, GloVe (Global Vectors for Word Representation) embeddings (pretrained on Common Crawl) were employed to capture semantic context. These 100-dimensional vectors were used to initialize the embedding layer in all deep learning models.

1. Tokenization: After text preprocessing, each word is mapped to a unique integer index based on the vocabulary. For example, ['this', 'game', 'is', 'stupid'] → [12, 45, 7, 89].
2. Embedding Layer: Indexed tokens are passed through a GloVe-initialized embedding layer, producing a matrix of shape ($sequence\ length \times 100$). This layer captures semantic similarity between words.
3. Model Input: The resulting embeddings serve as the input for sequential modelling using LSTM or Bi-LSTM architectures, allowing the model to learn contextual patterns over time.

2.3 Model Architecture

In this study, we divide the methodological process into two core components: the deep learning architecture and the training configuration. The architecture refers to the structural design of the neural network, including the types and sequence of layers (e.g., embedding, LSTM, dense output), their dimensions, and the flow of data through the model. In contrast, the training configuration focuses on how the model learns from data this includes the choice of

loss function, optimization algorithm, batch size, number of epochs, and training callbacks such as early stopping. While the architecture defines *what* the model is, the training setup defines *how* the model is optimized to perform the classification task effectively.

Two deep learning architectures were implemented and compared:

Table 3. Deep Learning Architectures

Layer	Output Shape	Description
Embedding	(150, 100)	GloVe embeddings (static)
LSTM / BiLSTM	(150, 128/256)	Contextual feature extraction
Global Max Pooling	(128/256)	Sequence compression

- **LSTM (Long Short-Term Memory):** A sequential model consisting of an embedding layer, a single LSTM layer, and a fully connected sigmoid-activated output layer for multi-label classification.
- **BiLSTM (Bidirectional LSTM):** A similar configuration as the LSTM model but with bidirectional processing to capture both forward and backward context in sentences.

All models used binary cross-entropy loss due to the multi-label nature of the classification problem. The Adam optimizer was employed for training.

2.3.1 Model Pseudocode Explanation and Gap

The preprocessed chat data used by the CNN and LSTM models is identical where each message is tokenized, then transformed into 100-dimensional GloVe embeddings and padded to a predetermined length of 150 tokens. The CNN-based model uses a global max-pooling layer to reduce the sequence to a single feature vector after these embeddings are run through a 1D convolutional layer (kernel size = 5) with ReLU activation to extract local n-gram features. Prior to a sigmoid (binary) or softmax (multi-class) output layer, this vector is fed through a dropout-regularized dense layer. The Adam optimizer with binary cross-entropy loss is used for training, and a held-out test split is used to assess performance.

The LSTM-based model uses the same embedding input but replaces the convolutional stage with a single LSTM layer (128 units) that processes the sequence and outputs its final hidden state. A global max-pooling is applied to that state to produce a fixed-length vector, which similarly flows through dropout, a dense layer, and the final activation. It is also trained with Adam and binary cross-entropy, then evaluated on the same test data

2.4 Evaluation Metrics

This study evaluates model performance using metrics tailored to multi-label classification: Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix.

1. Precision

Measures how many predicted toxic comments were actually toxic, useful when false positives (e.g., wrongly flagging non-toxic speech) are costly.

$$Precision = \frac{TP}{TP + FP} \quad (1)$$

2. Recall

Measures how many actual toxic comments were correctly identified. useful when false negatives (e.g., missing a threat) are more dangerous.

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

3. F1 – Score

A single metric that balances precision and recall by calculating their weighted average, providing a robust measure of a model’s overall accuracy in classification tasks.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

4. Macro – Averaged ROC – AUC

The Area Under the Receiver Operating Characteristic Curve averaged across all labels, treating each label equally regardless of its support. Calculates the AUC for each class and averages them equally, giving equal weight to rare classes.

5. Confusion Matrix

Shows breakdown of predictions vs ground truth for each label.

Application in the Study

1. Evaluation was performed per label, not as a single aggregate score, due to class imbalance (e.g., *threat* and *identity_hate* were underrepresented).
2. Results from both LSTM and BiLSTM models were compared using these metrics.
3. All metrics were computed using Scikit-learn’s multi-label evaluation functions.

3. Results

Table 4. Model Summary Table

Metric	Class 0 (Non-Toxic)	Class 1 (Toxic)	Class 2 (Mild-Toxic)	Overall Accuracy
LSTM Model	P: 0.53, R: 1.00, F1: 0.70	P: 0.00, R: 0.00, F1: 0.00	P: 0.00, R: 0.00, F1: 0.00	53.4%
CNN Model	P: 0.87, R: 0.96, F1: 0.91	P: 0.64, R: 0.65, F1: 0.64	P: 0.75, R: 0.60, F1: 0.66	79.9%

3.1 Model 1: LSTM-Based

This model used two stacked LSTM layers followed by a dense output layer. On paper, it achieved a final test accuracy of 53.4%. However, a closer look revealed that the model completely failed to learn meaningful patterns beyond the majority class. Specifically, it only predicted Class 0 (non-toxic), achieving a perfect recall of 1.00 but with mediocre precision (0.53) and an F1-score of 0.70. Both Class 1 (toxic) and Class 2 (severe-toxic) had precision, recall, and F1-scores of 0.00, showing that the model ignored these minority classes entirely. This outcome aligns with the significant class imbalance present in the dataset (as shown in Table 2), where the non-toxic class has more than twice the instances of the other classes combined. Such imbalance prevents the LSTM from capturing rare class signals, especially since its sequential memory mechanisms are not well-suited for short, imbalanced text data. This reinforces the importance of addressing data imbalance when using LSTM-based architectures for toxicity classification.

3.2 Model 2: CNN-based (Conv1D + GlobalMaxPooling)

For this model, we implemented a 1D convolutional architecture with global max pooling and dense layers. Unlike the LSTM, the CNN achieved robust and balanced results across all classes, with an overall test accuracy of 79.9%. It performed exceptionally well on Class 0 (non-toxic), recording precision 0.87, recall 0.96, and an F1-score of 0.91. More importantly, it successfully recognized and classified the minority classes by achieving F1-scores of 0.64 for Class 1 (toxic) and 0.66 for Class 2 (severe-toxic). Despite the data imbalance highlighted in Table 2, the CNN’s convolutional filters effectively captured localized n-gram patterns typical of toxic language in short gaming chat messages. These findings underscore the CNN’s advantage in handling class imbalance scenarios and support

its use as a more reliable baseline for multi-class toxicity classification. This performance further aligns with prior research showing CNNs' ability to extract discriminative local features more effectively than LSTM-based models.

4. Discussions

The results underscore several key insights into deep learning model performance for toxicity detection in gaming chat data. First, the LSTM model's collapse to predicting only the majority class (non-toxic) reflects its limitations in handling extreme class imbalance. This outcome highlights that LSTM-based sequential architectures, while powerful in longer-form text, are not well-equipped to capture rare patterns in short, highly imbalanced gaming chats. The data imbalance in the dataset (Table 2) clearly contributed to this performance collapse, as the LSTM was unable to extract features for the minority classes.

Second, the CNN model significantly outperformed the LSTM across all classes, with an overall test accuracy of 79.9% and meaningful F1-scores for minority classes. This performance aligns with previous research on short-text classification, where CNNs have proven effective in capturing local n-gram features. Despite the dataset's skew towards the non-toxic class, the CNN's convolutional layers successfully extracted discriminative patterns that signalled toxic language, even in sparse and brief messages typical of gaming chats.

Still, there are areas for improvement. In the context of toxic chat detection in online games, a false positive refers to a message that is non-toxic but incorrectly flagged as toxic by the model. For example, sarcastic remarks or competitive banter might be misclassified despite being harmless. Conversely, a false negative occurs when truly toxic messages, such as those containing personal insults, discriminatory language, or threats being mistakenly labelled as safe. These errors carry important implications in gaming environments, where chat is fast paced, emotionally charged, and often informal. Misinterpreting tone or slang can lead to unfair punishments or undetected abuse. False positives can frustrate players who feel wrongly muted or penalized, discouraging team communication and diminishing trust in the game's moderation system. Meanwhile, false negatives allow harmful content to persist, risking player safety, enabling harassment, and reinforcing toxic norms.

In our findings, the LSTM model exhibited a significant false negative tendency, failing to detect toxic or severe-toxic messages due to its overreliance on the majority class. The CNN model demonstrated stronger overall balance, successfully identifying many toxic cases and achieving reasonable F1-scores, though it still showed lower recall for the severe-toxic class. This suggests that even with more robust architectures, edge-case toxicity remains a detection challenges, particularly in skewed datasets. Addressing this will require techniques such as class-aware loss functions, threshold tuning, or data augmentation strategies that enhance model sensitivity to rare but critical toxic behaviours.

5. Conclusions

This study evaluated the effectiveness of CNN and LSTM neural network models in detecting toxicity in gaming chat data. Our results highlight the challenges of class imbalance, with the LSTM model failing to capture minority classes, while the CNN model demonstrated a more balanced and robust performance. Despite these advances, challenges remain in recognizing the most severe forms of toxicity. Future work will focus on integrating data balancing techniques and transformer-based architectures to further improve detection accuracy and reliability. Ultimately, the insights from this research can inform the development of real-world moderation tools for competitive online gaming environments, supporting healthier and more inclusive digital communities.

Data Availability

This study utilizes two publicly available datasets containing labeled in-game chat messages from Dota 2. The first, GOSU AI English Dota 2 Game Chats, is hosted on Kaggle and offers a large corpus of chat interactions labeled for toxicity, accessible at: <https://www.kaggle.com/danielfesalbon/gosu-ai-english-dota-chat/data>. The second, Dota 2 Toxic Chat Data, is available on Hugging Face and provides a curated, balanced set of toxic and non-toxic messages: <https://huggingface.co/datasets/dffesalbon/dota-2-toxic-chat-data>.

These datasets were merged and preprocessed for model training and evaluation in this study. All data is open-

access and freely usable for academic research. Preprocessing scripts can be provided upon request to the corresponding author.

Author Contributions

Kasandika Andariefli: Conceptualization*, Model Development, Writing – original draft. Jervino Leonard: Data Preprocessing, Validation, Writing – review & editing. Vincentius Dewanto: Experimentation, Visualization, Formal Analysis. Andien Dwi Novika: Supervision, Project Administration, Writing – review & editing.

References

- [1] Reddy, N., Ram, N., Kumar, K. K., and Murthy, K. S. R. (2022) “Toxic comments classification,” *International Journal of Recent Advances in Science Engineering and Technology* 10. <https://doi.org/10.22214/ijraset.2022.44500>.
- [2] Kou, Y. (2020) “Toxic behaviors in team-based competitive gaming: The case of League of Legends,” in *Proceedings of the ACM SIGCHI Annual Symposium on Computer-Human Interaction in Play*, New York, NY, pp. 81–92. <https://doi.org/10.1145/3410404.3414243>.
- [3] Maharani, A., Puspita, V., Aurora, R. A., and Wiranito, N. (2024) “Understanding toxicity in online gaming: A focus on communication-based behaviours towards female players in Valorant,” *Journal of Syntax Admiration* 5(5): 1559–1567. <https://doi.org/10.46799/JSA.V5I5.1137>.
- [4] Laato, S., Kordyaka, B., and Hamari, J. (2024) “Traumatizing or just annoying? Unveiling the spectrum of gamer toxicity in the StarCraft II community,” in *CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3613904.3642137>.
- [5] Frommel, J., Johnson, D., and Mandryk, R. L. (2023) “How perceived toxicity of gaming communities is associated with social capital, satisfaction of relatedness, and loneliness,” *Computers in Human Behavior Reports* 10: 100302. <https://doi.org/10.1016/j.chbr.2023.100302>.
- [6] Kordyaka, B., Laato, S., Weber, S., and Niehaves, B. (2023) “What constitutes victims of toxicity: Identifying drivers of toxic victimhood in multiplayer online battle arena games,” *Frontiers in Psychology* 14: 1193172. <https://doi.org/10.3389/fpsyg.2023.1193172>.
- [7] Beres, N. A., Frommel, J., Reid, E., Mandryk, R. L., and Klarkowski, M. (2021) “Don’t you know that you’re toxic: Normalization of toxicity in online gaming,” in *CHI Conference on Human Factors in Computing Systems*, May. <https://doi.org/10.1145/3411764.3445157>.
- [8] Sherman, J., and McClard, A. P. (2020) “Toxicity v. toxicity: How ethnography can inform scalable technical solutions,” *Ethnographic Praxis in Industry Conference Proceedings* 1: 279–297. <https://doi.org/10.1111/epic.12037>.
- [9] Sarker, J., Turzo, A. K., and Bosu, A. (2020) “A benchmark study of the contemporary toxicity detectors on software engineering interactions,” in *Asia-Pacific Software Engineering Conference*, Dec., pp. 218–227. <https://doi.org/10.1109/APSEC51365.2020.00030>.
- [10] Xia, Y., Zhu, H., Lu, T., Zhang, P., and Gu, N. (2020) “Exploring antecedents and consequences of toxicity in online discussions,” *Proceedings of the ACM on Human-Computer Interaction* 4(CSCW2). <https://doi.org/10.1145/3415179>.
- [11] Reid, E., Mandryk, R. L., Beres, N. A., Klarkowski, M., and Frommel, J. (2022) “Bad vibrations: Sensing toxicity from in-game audio features,” *IEEE Transactions on Games* 14(4): 558–568. <https://doi.org/10.1109/TG.2022.3176849>.
- [12] Ekiciler, A., Ahioglu, İ., Yildirim, N., Ajas, İ. İ., and Kaya, T. (2022) “The bullying game: Sexism based toxic language analysis on online games chat logs by text mining,” *Journal of International Women’s Studies* 24(3): 1–16. Available from: <https://www.proquest.com/openview/7e6d1aa29b949c7cd7c81bee509ca9aa> (Accessed: 7 July 2025).
- [13] Gupta, A., Nayyar, A., Arora, S., and Jain, R. (2020) “Detection and classification of toxic comments using LSTM and Bi-LSTM approach,” *Communications in Computer and Information Science* 1393: 100–112. https://doi.org/10.1007/978-981-16-3660-8_10.
- [14] Maity, A., More, R., Patil, A., Oza, J., and Kambli, G. (2024) “Toxic comment detection using bidirectional sequence classifiers,” in *2nd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)*, pp. 709–716. <https://doi.org/10.1109/IDCIOT59759.2024.10467922>.
- [15] Dubey, K., Nair, R., Khan, M. U., and Shaikh, P. S. (2020) “Toxic comment detection using LSTM,” in *International Conference on Advances in Electronics, Computing and Communication*, Dec. <https://doi.org/10.1109/ICAEECC50550.2020.9339521>.
- [16] Pavel, M. I., Razzak, R., Sengupta, K., Niloy, M. D. K., Muqith, M. B., and Tan, S. Y. (2021) “Toxic comment classification implementing CNN combining word embedding technique,” *Lecture Notes in Networks and Systems* 173: 897–909. https://doi.org/10.1007/978-981-33-4305-4_65.
- [17] Singh, I., Goyal, G., and Chandel, A. (2022) “AlexNet architecture-based convolutional neural network for toxic comments classification,” *Journal of King Saud University – Computer and Information Sciences* 34(9): 7547–7558. <https://doi.org/10.1016/j.jksuci.2022.06.007>.
- [18] Mishra, V., and Tripathi, M. (2022) “Detecting toxic comments using convolutional neural network approach,” in *14th IEEE International Conference on Computational Intelligence and Communication Networks (CICN)*, pp. 252–255. <https://doi.org/10.1109/CICN56167.2022.10008301>.
- [19] Ait-Saada, M., Role, F., and Nadif, M. (2021) “How to leverage a multi-layered transformer language model for text clustering: An ensemble approach,” in *International Conference on Information and Knowledge Management (CIKM)*, pp. 2837–2841. <https://doi.org/10.1145/3459637.3482121>.

- [20] Gupta, R. (2024) “От двунаправленных кодировщиков до новейших достижений: обзор BERT и его преобразующего влияния на обработку естественного языка,” *Informatics, Economics and Management* 3(1): 311–320. <https://doi.org/10.47813/2782-5280-2024-3-1-0311-0320>.
- [21] Laurer, M., Van Attevelde, W., Casas, A., and Welbers, K. (2024) “Less annotating, more classifying: Addressing the data scarcity issue of supervised machine learning with deep transfer learning and BERT-NLI,” *Political Analysis* 32(1): 84–100. <https://doi.org/10.1017/PAN.2023.20>.
- [22] Bongale, P., Razak, S., NCR, and RGM. (2021) “Transfer learning for spam text classification,” *International Journal of Recent Advances in Science Engineering and Technology* 9. <https://doi.org/10.22214/ijraset.2021.37349>.
- [23] Ngoc, T. V., Thi, M. N., and Thi, H. N. (2021) “Sentiment analysis of students’ reviews on online courses: A transfer learning method,” in *International Conference on Industrial Engineering and Operations Management*, pp. 306–314. <https://doi.org/10.46254/AP01.20210122>.
- [24] Wang, Z. (2023) “A new computationally efficient method to tune BERT networks – Transfer learning,” *Journal of Physics: Conference Series* 2580(1): 012012. <https://doi.org/10.1088/1742-6596/2580/1/012012>.
- [25] Garlapati, A., Malisetty, N., and Narayanan, G. (2022) “Classification of toxicity in comments using NLP and LSTM,” in *8th International Conference on Advanced Computing and Communication Systems (ICACCS)*, pp. 16–21. <https://doi.org/10.1109/ICACCS54159.2022.9785067>.
- [26] Diana, M. (2023) *Benchmarking AI techniques for toxicity detection in online games* [master’s thesis]. Utrecht (NL): Utrecht University. Available from: https://studenttheses.uu.nl/bitstream/handle/20.500.12932/44015/Thesis_MatteoDiana_8255083.pdf.
- [27] Merwick, F., Bi, Y., and Nicholl, P. (2021) “Performance evaluation of multi-class sentiment classification using deep neural network models optimised for binary classification,” in *Knowledge Science, Engineering and Management*, 12816: 624–635. https://doi.org/10.1007/978-3-030-82147-0_51.
- [28] Ghosal, D., Majumder, N., Mihalcea, R., and Poria, S. (2022) “Two is better than many? Binary classification as an effective approach to multi-choice question answering,” in *Conference on Empirical Methods in Natural Language Processing (EMNLP)*. <https://doi.org/10.48550/arXiv.2210.16495>.
- [29] Jamshidi, S., et al. (2024) “Effective text classification using BERT, MTM LSTM, and DT,” *Data & Knowledge Engineering* 151. <https://doi.org/10.1016/j.datak.2024.102306>.
- [30] Kaur, K., and Kaur, P. (2023) “Improving BERT model for requirements classification by bidirectional LSTM-CNN deep model,” *Computers and Electrical Engineering* 108. <https://doi.org/10.1016/j.compeleceng.2023.108699>.
- [31] Fesalbon, D. (2025) *GOSU AI English Dota 2 game chats* [dataset]. Kaggle. Available from: <https://www.kaggle.com/datasets/daniefesalbon/gosu-ai-english-dota-chat/data> (Accessed: 23 April 2025).
- [32] Fesalbon, D. (2025) *Dota 2 toxic chat data* [dataset]. Hugging Face. Available from: <https://huggingface.co/datasets/dfesalbon/dota-2-toxic-chat-data> (Accessed: 23 April 2025).